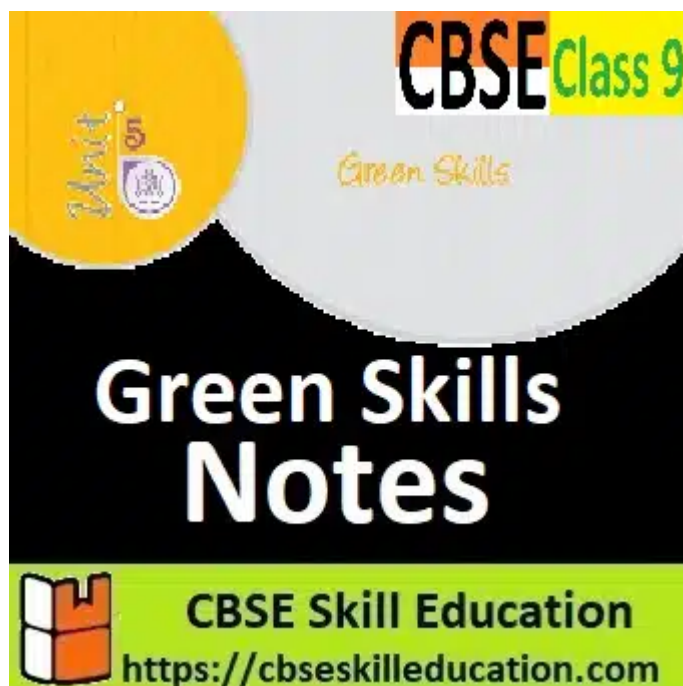


Green Skills Class 9 Notes

PART A - EMPLOYABILITY SKILLS | UNIT 05 GREEN SKILLS | NOTES

Source: <https://cbse-skillededucation.com/green-skills-class-9-notes/>

Green Skills Class 9 Notes help students understand the importance of protecting the environment and using natural resources responsibly. In today's world, environmental issues like climate change, pollution, and resource depletion are major concerns, making green skills essential for a sustainable future.



The environment influences every aspect of our lives, and our daily actions also have an impact on the environment. **We have to plan and maintain the environment in a sustainable manner so that we can enjoy the good environment created by us.**

Session 1: Society and Environment

What is the relation between society and environment?

Society affects the ecological balance of the environment due to the increase of the population, economic activities, people's involvement with nature, etc. The environment is also affected by the intensive agriculture and use of large amounts of natural resources like water, petroleum, minerals, wood, etc.

Natural resources

Humans use natural resources that are derived from nature. There are many ways in which natural resources can be found. They may be a solid, liquid, or gas. For example, water, soils, land, rocks, animals, forests, and fossil fuels.

- **Land Resources** – Land resources are used for production and residence. It is a finite resource that can be used for agricultural and infrastructure development.
- **Forest Resources** – Forest resources are used for making furniture, railway sleepers, boats, matches, etc.

- **Water Resources** – Water covers three-quarters of the earth’s surface and is a necessary element for life. Water resources include oceans, lakes, rivers, underground water, etc.
- **Mineral Resources** – Mineral resources are non-renewable and include metals like iron, copper, and aluminum and non-metals like salt, gypsum, clay, sand, and phosphates.
- **Food Resources** – Resources that provide food for organisms are called food resources. Plants are the source of food for herbivores and omnivores. Animals and birds are the source of food for carnivores and omnivores.

Natural resources fall under the following main categories:

- **Inexhaustible Resources** – The resources that cannot be finished by human consumption are called inexhaustible resources. For example, waterpower, wind power, solar energy, etc.
- **Exhaustible Resources** – The resources that are in limited quantities and finished if the continuous use is there. For example, coal, petroleum, etc.
- **Renewable Resources** – The resources that are constantly available or cannot be replaced or recovered. For example, forest trees, vegetative lands, etc.
- **Non-renewable Resources** – The resources that cannot easily be replaced once they are destroyed. For example, fossil fuels, minerals, etc.

How the human activities are damaging our earth and environment:

- **Overexploitation** – Harvesting a species from its habitat at a faster rate than the population can recover is known as overexploitation. For example, wild medicinal plants, destruction of forests, over hunting, overfishing etc.
- **Mining** – Many companies are digging below the earth’s surface to get ores. These ores are refined and extracted to valuable elements. For example, gems, metals, and minerals.
- **Deforestation** – when land is converted to a non-forest use such as construction or agriculture is known as deforestation.
- **Pollution** – The pollution derived from the Latin word “polluere,” meaning “to soil” or “defile (contaminate).” Pollution can be solid, liquid, or gaseous in nature. Pollution has harmful effects on animals, plants, and human beings.

Pollutants are of two types:

Pollutants that can decompose	Pollutants that cannot decompose
Pollutants that break down into simpler, harmless substances naturally (by the action of moisture and bacteria) are called biodegradable pollutants	Pollutants which cannot be broken down into simpler and harmless substances are called non-biodegradable pollutants. The harmful effect caused by these pollutants will be there for hundreds of years.
Examples: vegetable waste, sewage waste, paper, wood, cattle dung, agricultural waste from organic farms, etc.	Examples: plastics, insecticides, pesticides, chemicals like mercury or lead, aluminium, glass, etc.

Types of Pollution:

There are basically three types of pollution:

- **Land Pollution** – Damage to the land because of harmful substances is known as land pollution.

- **Water Pollution** – Adding harmful substances and disease-causing bacteria and other microorganisms to rivers, lakes, and oceans results in water pollution.
- **Air Pollution** – Addition of harmful gases and particles in air results in air pollution.

What is the greenhouse effect?

When the gases in the earth's atmosphere trap heat from the sun, this makes the earth warmer; this is known as the greenhouse effect.

What is climate change?

Climate change is a change in atmospheric temperatures and weather patterns. The earth is warming due to the greenhouse effect; some of the greenhouse gases enter the atmosphere due to burning fossil fuels like petrol, coal, and diesel. This greenhouse effect traps heat in the atmosphere, which increases the global warming.

What do you mean by Ozone Layer?

The atmosphere protects us from the harmful radiation from the sun. This atmosphere protection is done by using the 'Ozone Layer,' which is made of a gas called 'Ozone.' Refrigerators, air conditioners, and cleaning chemicals release ozone-depleting substances, such as chlorofluorocarbons, into the atmosphere.

Saving the environment: What can you do?

To save our environment, we need to –

- **Learning about the environment** – Learning about the environment – understanding our surroundings and related issues.
- **Learning through the environment** – Engaging with the environment inside and outside the classroom.
- **Learning for the environment** – Become responsible for the environment.

Reduce, Reuse, Recycle

There are three Rs which you can apply for saving the environment – Reduce, Reuse and Recycle. It is a concept of the modern waste management.

- **Reduce** – reduce the use of unnecessary items while shopping, buy items with minimal packaging, avoid buying disposable items and also avoid asking for plastic carry bags.
- **Reuse** – Reuse the materials for other purposes, such as making pillow covers or rags out of used shirts or ladies suits.
- **Recycling** – Recycling is reusing some components of the waste that may have some economic value. Some materials, such as aluminum and steel can be recycled many times. Metal, paper, glass and plastics are recyclable. Plastic items are recycled into new plastic products.

Session 2: Conserving Natural Resources

Conservation is the proper management of a natural resource to prevent its exploitation, destruction or degradation.

Soil conservation

Soil conservation means checking soil erosion and improving soil fertility by adopting various methods.

- **Maintenance of soil fertility** – The fertility can be maintained by adding manure and fertilizers.

- **Control on grazing** – Grazing should be allowed only on specified areas.
- **Reforestation** – Planting of trees and vegetation reduces soil erosion.
- **Terracing** – Dividing a slope into several flat fields to control rapid run of water.

Water conservation

Conservation and management of water are essential for the survival of mankind, plants and animals.

- Growing vegetation in the catchment areas, which will hold water.
- Constructing dams and reservoirs to regulate supply of water to the fields
- Sewage should be treated and only the clear water should be released into the rivers.
- Industrial wastes should be treated.
- Judicious use of water in our day-to-day life.
- Rainwater harvesting should be done

Energy conservation

Conservation of resources or energy means saving them and using them efficiently.

- Switch off lights, fans, TV and other electrical items, when not in use
- Use LED bulb to save energy.
- Keep the bulbs and tubes clean.
- Use pressure cooker to save energy.
- Keep vessels covered with a lid during cooking.
- Electric items like air conditioners geysers, heaters and dryers use a lot of electrical power. Use them when necessary.

Food conservation

Food is also preserved through various methods to prevent harmful bacteria and other microorganisms.

Forest conservation

Forest conservation involves preserving current forests or establishing new ones at the prescribed levels by the government or local authority.

Session 3: Sustainable Development and Green Economy

What is sustainable development?

Sustainability is the development that satisfies the needs of the present without compromising the capacity of future generations, guaranteeing the balance between economic growth, care for the environment and social well-being.

Sustainable development includes the following:

- Reducing excessive use of resources and enhancing resource conservation.
- Recycling and reuse of waste materials.
- Scientific management of renewable resources, especially bio-resources.
- Planting more trees.
- Green grassy patches to be interspersed between concrete buildings.

- Using more environment friendly material or biodegradable material.
- Use environmental friendly technology.

Sustainable Development Goals

Sustainable Development Goals are a universal goal of action to end poverty, protect the planet and ensure that all people enjoy peace and prosperity.

The Sustainable Development Goals (SDGs) were launched at the United Nations Sustainable Development Summit in New York in September 2015, forming the 2030 Agenda for Sustainable Development. The 2030 Agenda for Sustainable Development and its 17 Sustainable Development Goals (SDGs) adopted by world leaders in 2015, , embody a road map for progress that is sustainable and leaves no one behind.

What is Green growth?

The concept of green growth aims at achieving economic growth that is socially inclusive and environmentally sustainable.

What is Green Economy?

The term 'Green Economy' was first coined in a 1989 report for the Government of the United Kingdom, UNEP has defined the green economy as "one that results in improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities.

Components of a Green Economy

- Renewable energy
- Green building
- Well-managed (Sustainable) transport
- Water management
- Waste management
- Land management

What is Green skills?

The skills used for promoting green economy are known as green skills. These skills are needed in areas similar to renewable energy, sewer water treatment, climate resilient cities, green construction, solid waste management,etc.

Some of the areas in which green skills contribute to the sustainable development are as follows:

- Using renewable energy (example, using solar power and wind energy)
- Water and waste management
- Rain water harvesting
- Conserving energy
- Reducing pollution

What are green jobs?

A 'green job' is employment in any industry that contributes to preserving or restoring environmental quality in that sector and allowing for sustainable development.

Some of the sectors, which have the potential for green jobs are as follows:

- Agriculture
- Construction
- Energy
- Forestry
- Manufacturing
- Retail
- Automotive and Transport
- Tourism and Hospitality
- Education

What is Green project?

Many people and organisation are concerned and motivated about doing something to save the environment. They are implementing green projects in areas like waste management, energy conservation, green sanitation, biofuel use, green buildings, etc.